

Flood Preparedness AR Game

Weekly / Gantt-Style Development Timeline

This timeline aligns the 31-day development plan into a structured weekly format suitable for a capstone calendar. Weeks are task-focused and can be mapped directly to actual academic dates.

Week 1 – Pre-Production & Planning (Days 1–3)

- Finalize storyline based on storyboard
- Break story into missions and scenes
- Define game rules, scoring, and fail states
- Create question bank and Go-Bag item list
- Define achievement conditions

Week 2 – Unity Setup & Core Framework (Days 4–8)

- Create Unity LTS mobile project
- Install AR Foundation, ARCore, ARKit
- Configure Android/iOS build settings
- Create folder structure
- Implement scene navigation system
- Implement basic game state manager

Week 3 – Dialogue & Q&A; Systems (Days 9–13)

- Dialogue UI (character portrait + text box)
- Typewriter text effect (optional)
- Question & Answer UI
- Answer validation logic
- Correct/incorrect feedback panels
- Safety meter or score tracking

Week 4 – Inventory & Go-Bag Logic (Days 14–17)

- Create ItemData ScriptableObjects
- Define essential vs non-essential items
- Backpack capacity rules
- Backpack UI and packed item list
- Mission validation for required items

Week 5 – AR Foundation Base & Item Spawning (Days 18–22)

- Create AR Scene
- AR Session and XR Origin setup
- Plane detection and raycasting

- AR item prefabs (collider + visuals)
- Mission-based item spawning logic
- Fallback spawn if no plane detected

Week 6 – AR Drag & Drop Interaction (Days 23–26)

- Touch selection via raycast
- Drag movement logic
- Drop detection into Backpack UI
- Item collection animation & sound
- Update inventory and checklist in real time

Week 7 – Content Expansion & Polish (Days 27–30)

- Add remaining missions from storyboard
- Achievement popups and badges
- Tutorial prompts (scan, drag, drop)
- Sound effects and UI animations
- Performance optimization

Week 8 – Testing, Fixes & Final Build (Day 31)

- Full playthrough testing
- Device testing (Android / iOS)
- Edge-case testing (low light, no planes)
- Bug fixes
- Final APK/IPA build and demo recording

Gantt-Style Phase Summary

Week	Primary Focus	Days
Week 1	Planning & Design	1–3
Week 2	Unity Setup & Framework	4–8
Week 3	Dialogue & Q&A	9–13
Week 4	Inventory & Go-Bag	14–17
Week 5	AR Setup & Spawning	18–22
Week 6	AR Drag & Drop	23–26
Week 7	Content & Polish	27–30
Week 8	Testing & Deployment	31

This weekly structure is designed to map cleanly onto an academic capstone calendar. If specific start and end dates are provided, this timeline can be converted into a dated Gantt chart.